Event Contracts and the Liquidity Threshold

Risk Transfer, Sportsbook Exposure, and the Derivatives Case for Event Markets

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ABSTRACT

Event contracts are often described as speculative wagers. Under U.S. derivatives law, however, the relevant inquiry is not whether speculation exists, but whether a contract can facilitate legitimate risk transfer or price discovery in underlying economic activity. This paper examines whether event contracts tied to sports outcomes could serve that function for sportsbooks, which routinely warehouse asymmetric event-outcome exposure. It introduces a liquidity-aware framework—the Liquidity-Constrained Event Risk Transfer Curve—to illustrate how the hedgeable fraction of sportsbook exposure depends on market depth. The paper's core claim: **prediction markets do not need commercial hedgers to demonstrate economic purpose; they need liquidity sufficient to enable hedging.**

KEY FINDINGS

- Derivatives regulation historically tolerates speculation because it facilitates risk transfer and price discovery.
- Prediction markets challenge the regulatory boundary between derivatives law and gambling law.
- The economic function of event contracts depends critically on liquidity.
- Early markets appear speculative because commercial hedging is infeasible at shallow depth.
- Once liquidity crosses a threshold, event contracts can facilitate real and material risk transfer.
- The absence of current commercial hedging does not establish lack of economic purpose—it may reflect the liquidity constraints typical of early-stage derivatives markets.
- Prediction markets may represent the financialization of event risk.

EXECUTIVE SUMMARY

The central policy insight of this paper is this: **it is analytically unsound to treat the absence of early commercial hedging as conclusive proof that an event contract lacks economic purpose.** In many derivatives markets, commercial hedgers are among the last major participants to arrive. Evaluating event contracts solely by present participation risks confusing market infancy with structural incapacity.

Debate over event contracts frequently begins with the wrong question. The issue is often framed as whether a contract "looks like gambling." But that framing misses the analytical center of derivatives regulation. The more relevant question is whether a contract can redistribute real economic risk or assist in pricing it.

Sportsbooks provide a concrete use case. Because books are rarely perfectly balanced, bookmakers routinely accumulate event-outcome exposure. Existing tools—odds moves, betting limits, and layoff arrangements—can mitigate but do not eliminate that exposure. Event contracts could, in principle, provide an additional market-based channel through which that risk is distributed to speculators, arbitrageurs, market makers, and eventually institutional counterparties willing to assume it.

This paper offers an illustrative, liquidity-aware model rather than an empirically calibrated sportsbook valuation model. Its purpose is to clarify market structure. The model's principal output is the Liquidity-Constrained Event Risk Transfer Curve, which shows how the hedgeable fraction of sportsbook exposure declines when liabilities grow faster than available event-market depth. Whether hedging is observed today is different from whether hedging could emerge once liquidity matures.

I. INTRODUCTION

Event contracts—derivatives that settle based on the occurrence of discrete events—have become central to current debates over the proper scope of regulated derivatives markets. Critics often characterize such contracts as wagers. But nearly every derivatives market contains speculative participation. The better question is whether a contract can facilitate risk transfer or price discovery in a way that is economically meaningful.[1]

One setting in which the risk-transfer question becomes concrete is sports betting. Sportsbooks do not merely post lines; they warehouse financial exposure to sporting outcomes. When betting flows skew sharply toward one side of an event, the bookmaker may face concentrated losses if that outcome occurs. This paper explores whether liquid event markets could allow part of that exposure to be transferred outward, in a manner analogous to more familiar derivatives markets.[2]

The paper builds on the author's earlier commentary in this series regarding the economic-purpose framework for prediction markets and the role of liquidity in institutional adoption.[3] It extends that work by supplying a compact conceptual model and figures intended to clarify the relationship among liquidity, hedge capacity, loss-distribution improvement, and regulatory analysis. Interactive versions of the core figures are available at eventrisk.ai.

The central question addressed here is not whether commercial hedging is presently observed in event markets, but whether the structure of such markets could support meaningful risk transfer once sufficient liquidity develops.

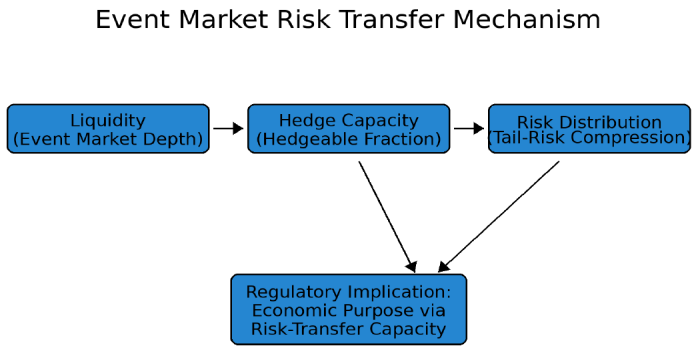

Figure 0 — Event Market Risk Transfer Mechanism.

Greater event-market liquidity expands hedge capacity; greater hedge capacity compresses the left tail of sportsbook outcomes; and that relationship informs the legal question whether event contracts can serve an economic purpose through risk-transfer capacity.

II. REGULATORY FRAMEWORK FOR EVENT CONTRACTS

The Commodity Futures Trading Commission has historically evaluated novel derivatives contracts through the lens of economic function. Courts applied what was often described as an economic purpose test, examining whether a contract could facilitate legitimate hedging or price discovery in an underlying market.[1] That framework does not require that commercial hedging already occur; it asks whether the structure of the contract and the surrounding market could, under realistic conditions, support such activity.

Event contracts listed on regulated derivatives exchanges are not evaluated in a vacuum. Designated contract markets operate within a statutory framework that includes core principles relating to market integrity, surveillance, position management, orderly trading, and related protections.[5] That framework does not require a new contract to arrive with commercial hedgers already active on day one.

The current event-contracts backdrop—including Regulation 40.11 and the ongoing debate surrounding contracts tied to gaming-related outcomes—underscores why clear market-structure analysis is essential.[6]

A parallel line of analysis developed in recent litigation commentary suggests that CFTC jurisdiction may attach on an independent basis: exchange-traded sports event contracts are now arguably "commonly known to the trade as a swap" within the meaning of CEA § 1a(47)(A)(iv), given that multiple designated contract markets list them as swaps, futures commission merchants offer them as swaps, and derivatives clearing organizations clear each trade accordingly.[9] On that reading, the "event or contingency associated with a potential financial, economic, or commercial consequence" subparagraph—which has occupied the courts in nationwide litigation—may not be the decisive provision at all, because the industry's own uniform terminology independently satisfies the statute. The economic-purpose framework developed in this paper remains relevant regardless of which jurisdictional pathway applies. Once CFTC jurisdiction attaches, the Commission's core-principles review under CEA § 5(d) requires an assessment of whether the contract serves the price-discovery and risk-transfer functions that justify regulated derivatives markets.[5] This paper's liquidity-threshold analysis speaks directly to that substantive inquiry.

III. RELATED LITERATURE

A first strand of literature concerns derivatives hedging theory. Holbrook Working's classic treatment emphasized that hedging should not be reduced to a simple story of risk elimination; rather, hedging allows actors exposed to an underlying risk to transfer part of that exposure to others who are willing to bear it.[1] That perspective remains useful because it focuses attention on function rather than form.

A second strand concerns information aggregation in prediction markets. Wolfers and Zitzewitz show that prediction markets can aggregate dispersed information about uncertain future events and, under

appropriate conditions, generate useful forecasts.[4] Most of that literature emphasizes predictive efficiency. Comparatively less attention has been paid to a distinct question: when these same markets might also facilitate risk transfer for commercial actors exposed to the underlying event.

A third strand concerns market structure. Commercial hedgers rarely inaugurate a derivatives market. Speculators and market makers typically arrive first, generating the depth needed for later institutional and commercial participation. That sequencing is central here because a market can possess risk-transfer capacity before large hedgers visibly use it.

The most direct scholarly challenge to the economic-purpose framework as applied to event contracts is Beylin (2025), which argues that the CFTC's twin statutory goals—hedging and price discovery in underlying "cash" markets—operate as a limiting principle that many prediction market contracts fail to satisfy.[7] Beylin's analysis is careful and the limiting-principle insight is well-taken. But the argument, applied to sports-related event contracts, relies on a static assessment of current market participants rather than a forward-looking inquiry into whether a developing market is capable of supporting risk transfer as liquidity matures. The liquidity-threshold framework developed here directly answers that concern: it explains why commercial hedging is sparse in early-stage markets (depth is insufficient) and identifies the conditions under which the Beylin limiting principle would be satisfied rather than violated. A contract tied to a sporting outcome on which a sportsbook holds concentrated exposure is not analytically equivalent to a contract on a Taylor Swift album chart position—the former connects to real commercial risk warehoused by an identifiable class of potential hedgers; the latter does not. Lockhart (2025) likewise examines the legal boundaries of speculative event markets and provides a complementary treatment of where the gambling/derivatives line appropriately falls.[8]

IV. RISK WAREHOUSING IN SPORTS BETTING

Sportsbooks accept wagers on binary or near-binary outcomes: a team wins or loses; a point spread covers or fails; a total lands over or under. Perfectly balanced books are uncommon, particularly around marquee events. In practice, flows can cluster on one side of a market, leaving the bookmaker with significant contingent liability if that favored outcome occurs.

Consider a simplified championship-game example. If $250 million is wagered, with 65 percent of the stake on the favorite and 35 percent on the underdog, the bookmaker may face concentrated payout exposure if the favorite wins. The exact economics vary by pricing, hold, and mix of bets. The underlying point is more basic: sportsbooks warehouse event-driven risk that is economically real, not merely rhetorical.[2]

Illustrative Exposure Profile

Item	Value	Implication
Total wagers	$250 million	Large headline event
Favorite-side wagers	$162.5 million	Skewed book toward one outcome
Underdog-side wagers	$87.5 million	Insufficient natural offset
If favorite wins	—	Potential tail-loss event for the book

V. WHY SPORTSBOOKS ARE NATURAL COMMERCIAL HEDGERS

Among potential commercial participants in event markets, sportsbooks occupy a particularly clear position as natural hedgers. Their exposure arises directly from the same outcomes on which event contracts settle. In this respect, sportsbooks resemble airlines exposed to fuel prices, farmers exposed to crop prices, or banks exposed to rate movements: each holds a business position that can become economically painful if the underlying state of the world moves the wrong way.

The analogy should not be overstated. Sportsbooks already possess internal risk-management tools and may not hedge every exposure even in a mature market. But from a corporate-finance perspective, reducing severe event-outcome volatility could smooth earnings and reduce capital stress around marquee events. Research on comparable intermediary businesses—exchange operators, clearinghouses, and primary insurers—consistently shows that markets assign higher valuation multiples to earnings streams with lower variance, even where absolute margins compress. The more liquid the event market, the more plausible such hedging becomes.

VI. LIQUIDITY AND DERIVATIVES MARKET FORMATION

Commercial hedgers rarely appear first. A stylized market-development sequence:

Speculators → Market Makers → Institutional Liquidity → Commercial Hedgers

That sequence matters because regulatory analysis can go wrong if it demands visible commercial hedging at the inception of a market. Early trading often reflects directional speculation, basis views, or market-making activity. Those participants help create the very depth that later makes commercial risk transfer possible.

The feasibility of hedging therefore depends on two variables simultaneously: the size of the sportsbook's liability and the depth of the event market available to absorb offsetting positions. In shallow markets, price impact and limited order-book depth can prevent meaningful hedging even when a coherent hedging use case exists. Conversely, once markets reach sufficient depth, the same contracts can suddenly support economically meaningful risk transfer.

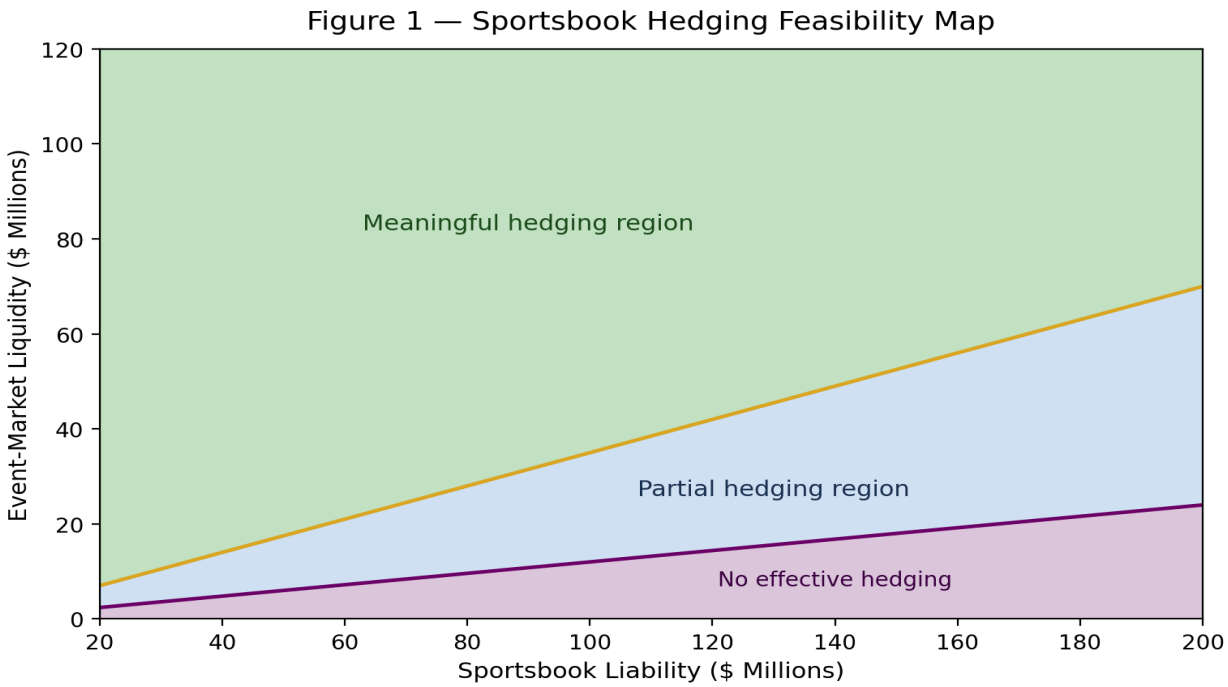

Figure 1 — Sportsbook Hedging Feasibility Map.

The map visualizes how hedging viability depends jointly on sportsbook liability and available event-market liquidity. Shallow markets may support little or no meaningful hedging for large liabilities; deeper markets move the same exposure into a partial or meaningful hedging region.

VII. LIQUIDITY THRESHOLD FOR COMMERCIAL HEDGING ADOPTION

Commercial hedging activity may remain sparse in early-stage event markets even where a coherent hedging use case exists. Practical hedging depends on a threshold level of liquidity below which institutional participation is uneconomic. Once depth crosses that threshold, however, adoption can accelerate: the same participants who rationally stayed out of thin markets find it feasible to hedge once sufficient counterparty depth exists.

VIII. SPORTSBOOK EXPOSURE VS. EVENT-MARKET HEDGING CAPACITY

The relationship between sportsbook exposure and event-market liquidity can be expressed as a market-structure map. The same exposure may be entirely unhedgeable in a thin market, partially hedgeable as depth develops, and meaningfully hedgeable once order-book depth becomes sufficient to absorb institutional-size positions without prohibitive price impact.

IX. THE LIQUIDITY-CONSTRAINED EVENT RISK TRANSFER CURVE

The paper’s central output is the Liquidity-Constrained Event Risk Transfer Curve. The curve does not attempt to replicate any single bookmaker’s exact economics. Instead, it shows how the fraction of exposure that can be hedged changes as liabilities grow relative to available market depth. In shallow

markets, only small exposures can be offset before price impact, slippage, or insufficient depth becomes binding. In deeper markets, larger fractions remain hedgeable for longer.

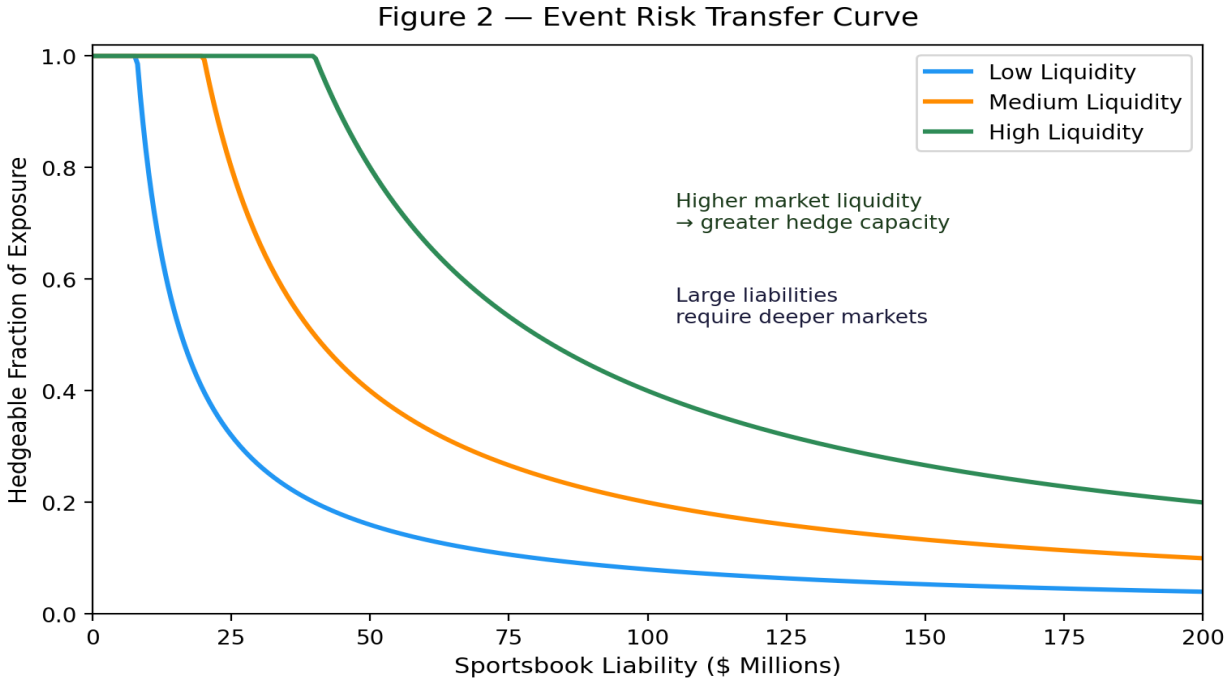

Figure 2 — Liquidity-Constrained Event Risk Transfer Curve.

The x-axis reports sportsbook liability; the y-axis reports the hedgeable fraction of that exposure. Separate curves represent low-, medium-, and high-liquidity regimes. The core implication: liquidity, not merely contractual design, governs practical hedge capacity.

X. SPORTSBOOK RISK PROFILE UNDER HEDGING

A contract may be theoretically hedgeable yet still economically unattractive. The next question is whether access to event-market hedges changes the sportsbook's risk profile in a way that matters. To make the comparison legible to non-specialists, the analysis uses **Expected Worst-Case Loss (95%)**: the average loss in the worst five percent of simulated outcomes.

Table 1 — Illustrative Hedging Scenarios

Scenario	Expected Value	Expected Worst-Case Loss (95%)	Maximum Loss
Unhedged	–$66M	–$120M	–$130M
Hedged — shallow liquidity	–$69M	–$118M	–$120M
Hedged — deep liquidity	–$67M	–$92M	–$88M

Deep-liquidity hedging reduces worst-case loss by roughly 27 percent relative to the unhedged baseline while preserving a broadly similar expected-value profile.

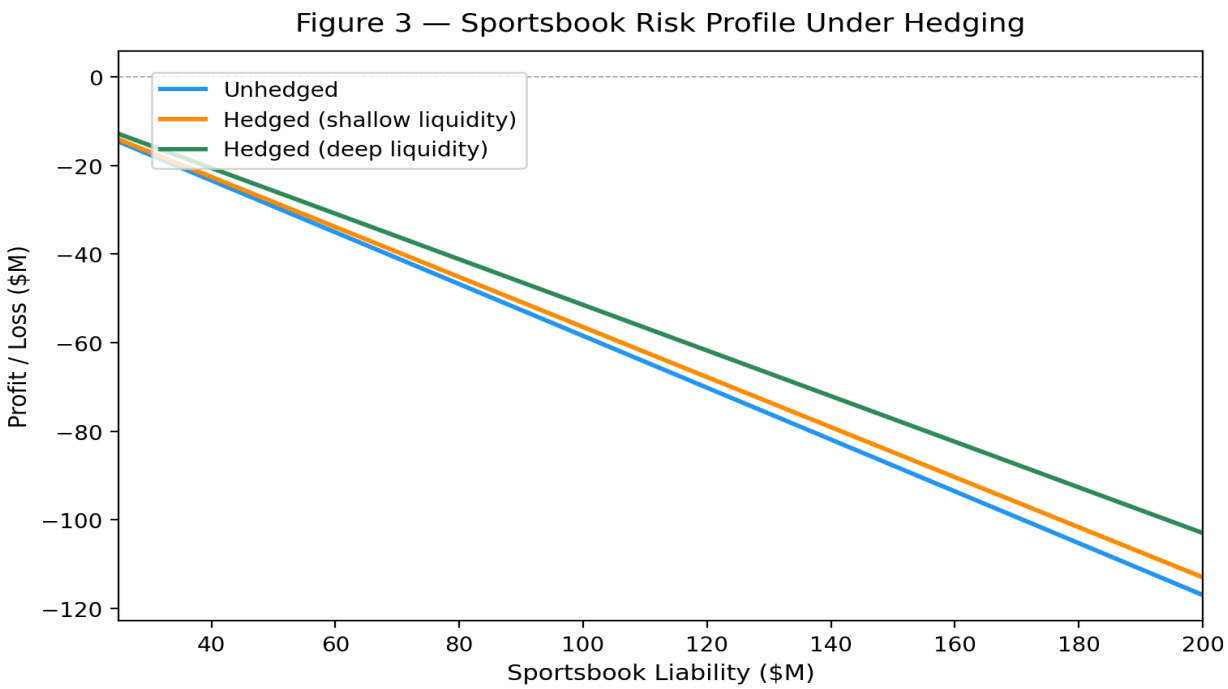

Figure 3 — Sportsbook Risk Profile Under Hedging.

The unhedged curve shows direct exposure as liability increases. The shallow-liquidity hedge offers only modest improvement. The deep-liquidity hedge compresses worst-case downside by roughly 27 percent while preserving a broadly similar expected-value profile—evidence that deep markets produce economically meaningful risk reduction.

XI. TAIL-RISK COMPRESSION

The distributional view is often the most intuitive. Even when average economics remain roughly similar, the left tail can improve meaningfully. That matters because catastrophic sportsbook losses are precisely the outcomes most likely to strain capital, create earnings volatility, or motivate the search for outside risk-transfer mechanisms.

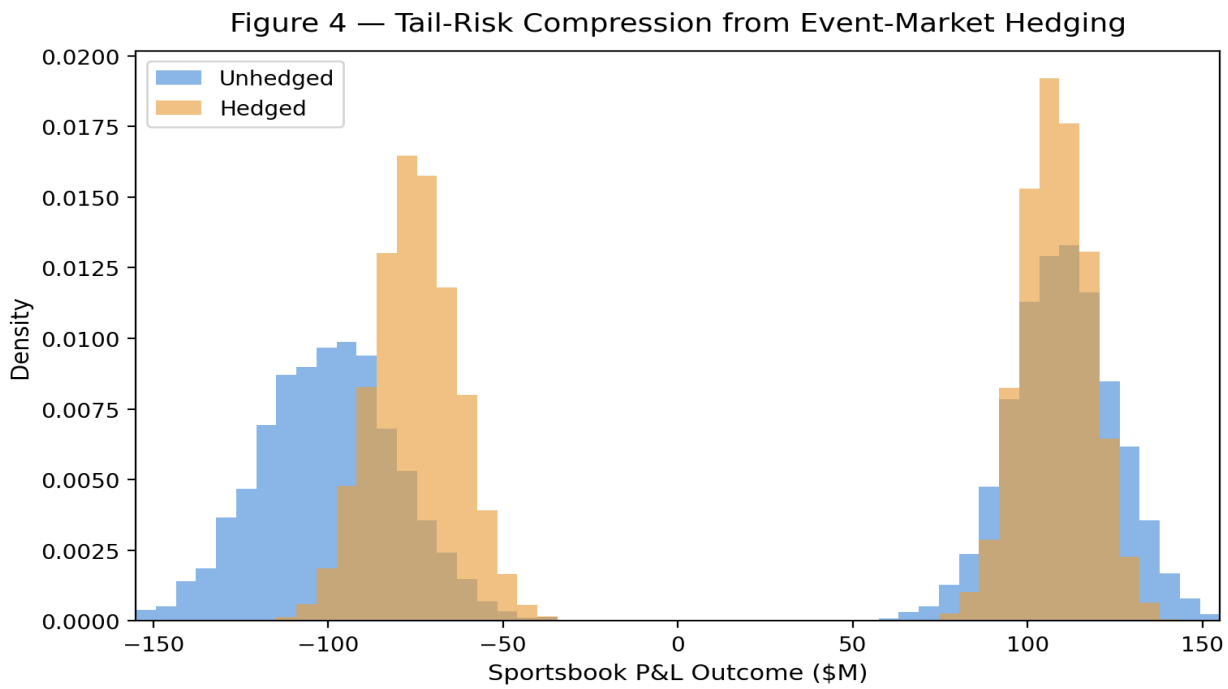

Figure 4 — Tail-Risk Compression from Event-Market Hedging.

The hedged distribution narrows and the left tail shifts inward, indicating fewer and less severe catastrophic-loss outcomes. The figure shows comparative shape, not a calibrated empirical density for any specific sportsbook.

XII. HEDGING EFFICIENCY FRONTIER

Hedging is not free. A sensible comparison therefore weighs tail-risk reduction against expected value sacrificed. The resulting frontier differs materially across liquidity regimes. In deep markets, the same expected-value concession buys more downside protection than in shallow markets—a result with direct implications for when commercial hedging becomes attractive.

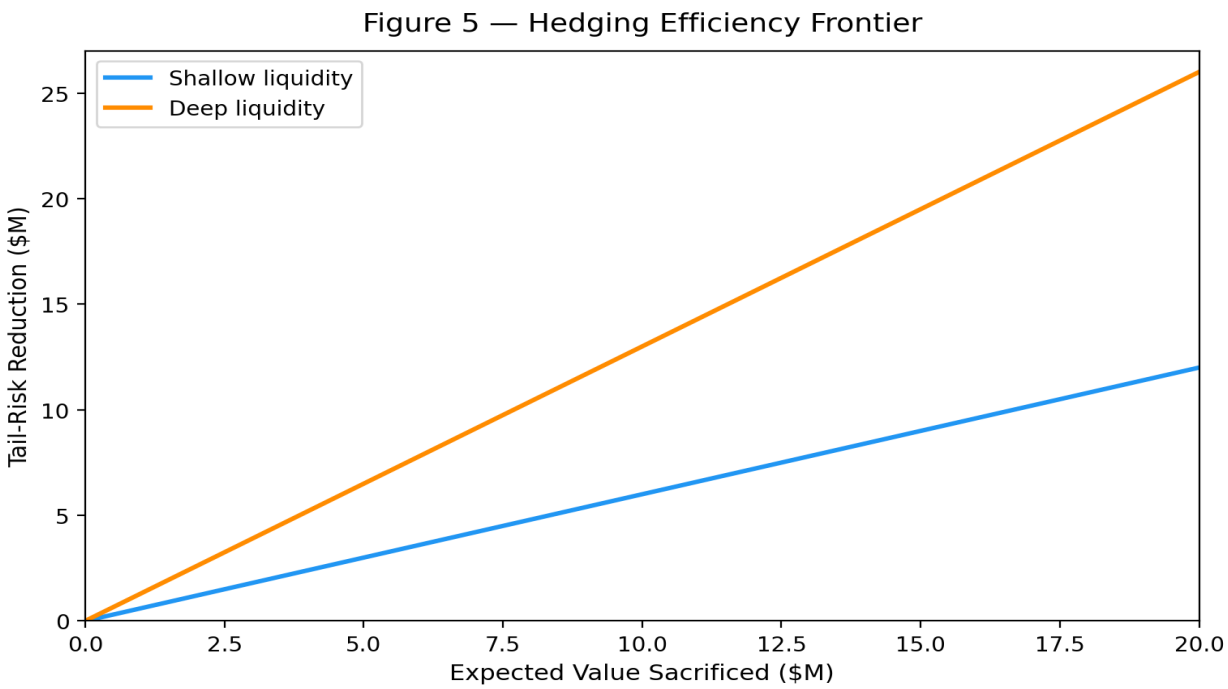

Figure 5 — Hedging Efficiency Frontier.

The deep-liquidity frontier lies outward from the shallow-liquidity frontier, indicating that better market depth improves the tradeoff between expected value retained and tail-risk reduced.

XIII. RISK-TRANSFER CAPACITY VS. OBSERVED HEDGING

This distinction is the paper's core legal and economic point. Regulatory debates often ask whether commercial actors currently hedge in an event market. But a market can possess *risk-transfer capacity* before such hedging is actually observed. Put differently: observed usage is evidence, but it is not the only possible evidence of economic function.

If market depth is insufficient, commercial hedgers may rationally stay out. Their absence in that circumstance says little about whether the contract would facilitate risk transfer in a mature market. Evaluating event contracts solely by present participation therefore risks confusing infancy with incapacity.

> *The relevant question is not simply, "Do sportsbooks hedge today?"*
> *It is: Would these contracts allow meaningful hedging if liquidity develops?*

Prediction markets do not prove economic purpose through usage alone. They prove it through potential risk-transfer capacity once a regulated market acquires sufficient depth to support economically meaningful hedging.

XIV. POLICY IMPLICATIONS FOR EVENT MARKET REGULATION

The analysis here does not claim that every sports-related event contract should be listed, nor that every sportsbook will hedge, nor that liquidity will automatically deepen to the point where meaningful hedging emerges. The claim is narrower and more useful: the economic-purpose inquiry should account for whether a market is capable of supporting risk transfer as liquidity evolves.[26]

That framing aligns the policy question with market mechanics. A regulator may still conclude that a specific contract is impermissible or undesirable for independent reasons. But it is analytically unsound to treat the absence of early commercial hedging as conclusive proof that a contract lacks economic purpose. In many derivatives markets, hedgers are among the last major participants to arrive. The sequencing of speculator-first, hedger-later is a feature of market development, not a defect in the underlying contract.

XV. TESTABLE PREDICTIONS

A framework with economic content should generate falsifiable predictions. This one does:

- Commercial hedging activity should be sparse in early-stage event markets, even where a coherent hedging use case exists. This is not evidence of incapacity; it is the expected pattern of market development.
- As event-market depth improves around marquee events, hedge sizes and hedging participation should become more economically plausible—and that increased plausibility should be observable in order-book depth and open interest.
- If sportsbooks eventually use event markets, the first meaningful adoption is more likely in highly liquid, high-visibility event clusters than in thinly traded long-tail markets.

XVI. CONCLUSION

Prediction markets do not need immediate commercial hedgers to demonstrate economic purpose; they need liquidity sufficient to make hedging economically viable. Once that distinction is made explicit, the policy debate changes. The relevant question becomes not whether event contracts superficially resemble wagers, but whether regulated event markets can, under the right liquidity conditions, redistribute real economic exposure.

For sportsbooks, that possibility is not abstract. They already warehouse event-outcome risk. If event contracts can transfer part of that risk to broader financial markets, they begin to look less like isolated wagers and more like the kind of risk-transfer instruments derivatives law has long recognized. The analogy to commodity futures, interest-rate swaps, and catastrophe bonds is imperfect—but the underlying function is the same: moving concentrated exposure from those who hold it by necessity to those who accept it by choice.

Whether event markets ultimately develop to that point is an empirical question. The legal question, properly framed, is whether they are capable of doing so. This paper argues they are—provided liquidity reaches the threshold at which commercial hedging becomes economically rational. That is the correct threshold for the economic-purpose inquiry, and applying it would bring prediction market regulation into alignment with how derivatives law has always worked.

NOTES AND AUTHORITIES

1. Holbrook Working, *Hedging Reconsidered*, 35 American Journal of Agricultural Economics 544 (1953).

2. Statement of Commissioner Dan M. Berkovitz Related to Review of ErisX Certification of NFL Futures Contracts (Apr. 7, 2021), Commodity Futures Trading Commission.

3. David Kuhn, *Prediction Markets and the Economic Purpose Test* (2026) (companion piece, Part I of this series), available at https://x.com/gr8day4this/status/2021331029009563896; David Kuhn, *The Financialization of Event Risk* (2026) (companion piece, Part II of this series), available at https://x.com/gr8day4this/status/2023875319996838148.

4. Justin Wolfers & Eric Zitzewitz, *Prediction Markets*, 18 Journal of Economic Perspectives 107 (2004).

5. Commodity Exchange Act § 5(d), 7 U.S.C. § 7(d) (core principles for designated contract markets).

6. 17 C.F.R. § 40.11; CFTC, Contracts & Products (describing event contracts and current Regulation 40.11 framework).

7. Ilya Beylin, *Event Contracts Are a Step Too Far for Derivatives Regulation*, 4 University of Chicago Business Law Review 77 (2025).

8. Karl Lockhart, *Betting on Everything* (Feb. 1, 2025), available at https://ssrn.com/abstract=5317376.

9. Rob Schwartz, *Federal Preemption in Sports Prediction Market Litigation: This Shouldn't Be a Jump Ball*, 46 Futures & Derivatives L. Rep. 3 (Mar. 2026). Schwartz served as General Counsel and Deputy General Counsel for Litigation at the CFTC before joining private practice.

Author's note: Interactive versions of the core figures are available at eventrisk.ai/explainer. All charts in this paper are illustrative and are intended to clarify comparative dynamics rather than to present empirically calibrated sportsbook financial forecasts.